

Referee package – STEP-5B rectangle-constant closure: the theorem-grade rectangle prefactor

TECT verification-first repository · claim B5-BEYOND-LAYER-BOUND

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1. Purpose and scope

This referee document concerns the STEP-5B rectangle-constant closure: the theorem-grade prefactor $K(n)$ controlling the layer budget. **Scope:** the rectangle constant and the official threshold. **Not claimed:** the alternative 20/9 incidence route is provisional (constant unpinned) and is not relied upon. This package does not by itself close full STEP-5B (the budget comparison and admissibility gates are tracked separately), does not close admissible-class exhaustiveness, and does not enact the full Reading-H C_{full} theorem.

2. Definitions and the result

Definitions (patch v1.1-1). Let $Q \subset S_{q_0}^2$ be an admissible pattern of n independent modes on the condensation shell, with symmetrised (antipodal) point set $\hat{Q} = Q \cup (-Q)$ of $N = 2n$ points and amplitudes $\{A_a\}$ of total intensity $I = \sum_a A_a^2$. The rectangle prefactor is the normalised off-diagonal transfer ℓ^2 mass

$$K(n) = \frac{1}{(\lambda' I)^2} \sum_{t \neq 0} w_t^2, \quad w_t = \lambda' \sum_{\substack{a+b=t \\ a,b \in \hat{Q}}} A_a A_b,$$

i.e. the quantity the STEP-5B layer budget must dominate. The balance parameter κ is the registered **H-KBAL** amplitude-balance constant,

$$\mathbf{H-KBAL}: \quad A_{\max}^2 \leq \kappa^2 I/n,$$

with $\kappa \simeq 1$ at the production operating point (equal-amplitude patterns have $\kappa = 1$ exactly).

The result. (i) **THEOREM** (operator verdicts #9/#10): for every κ -balanced admissible pattern,

$$K(n) \leq 8 + 4\sqrt{14} \kappa^4 \sqrt{n}, \quad c_R^{\text{rigorous}} = 4\sqrt{14} \simeq 14.97,$$

with 1.59×10^5 the **OFFICIAL** threshold (derivation in Sec. 3). (ii) The 20/9 incidence route (2.2×10^{10}) is provisional with an unpinned constant and is recorded but not load-bearing.

Provenance (patch v1.1-3). The rectangle theorem is imported from operator verdicts #9/#10, archived in the folder note **rectangle-constant-closure-260605-260605-v1.3.tex.txt** §2–§3: verdict #9 supplied the derivation $\sum_C p_C^3 \leq 7N^3$ (carrier-circle cube bound) with the Cauchy–Schwarz interpolation $\sum_C p_C^2 \leq (\sum_C p_C)^{1/2} (\sum_C p_C^3)^{1/2} \leq \sqrt{7} N^{5/2}$; verdict #10 repaired the provisional incidence exponent to 20/9 (correcting the 28/13 arithmetic slip). This package records and machine-reproduces the numerical threshold but does not reprove the incidence alternative.

3. Structure of the argument and threshold derivation

The rectangle prefactor is bounded by the closed-form $8 + 4\sqrt{14}\kappa^4\sqrt{n}$ (the operator-verified rectangle theorem). STEP-5B closes for a pattern of n modes provided (patch v1.1-2)

$$K(n) \leq 8 + 4\sqrt{14}\kappa^4\sqrt{n} < K_{\text{budget}}(I) = \frac{4(1 - a_0(I)) \Delta F_{\text{margin}}}{(\lambda' I)^2 J_{\text{max}}},$$

which at the production anchor $I = 4 \times 10^{-4}$ (where $K_{\text{budget}} = 5972$, $\kappa \simeq 1$) gives the official admissible-mode threshold

$$n_{\text{adm}} \lesssim \left(\frac{K_{\text{budget}} - 8}{4\sqrt{14}\kappa^4} \right)^2 = \left(\frac{5972 - 8}{4\sqrt{14}} \right)^2 = 1.59 \times 10^5.$$

(3.77×10^3 at $I = 10^{-3}$; 2.02×10^2 at $I = 2 \times 10^{-3}$.) The incidence-geometry alternative would relax this to 2.2×10^{10} but its constant is not pinned, so it is documented for cross-check only and is not used in the closure.

4. Numerical reproduction

The beyond-layer Gershgorin bound script reproduces the rectangle constant and the 1.59×10^5 threshold with asserts; run it from the bundle. Clean-run record (2026-06-12, inspection re-run): 192/192 asserts PASS, exit 0; key asserts `operator_p3_bound` (3 configs), `operator_interpolation` (3 configs), `operator_cr_rigorous` ($c_R = 14.9666$), `operator_theorem_region_anchor` ($n \lesssim 1.59 \times 10^5$ at $I = 4 \times 10^{-4}$), `incidence_exponent_repaired_20_9`, `incidence_route_region_repaired` (provisional, not load-bearing); JSON artefact refreshed at `claims/B5-BEYOND-LAYER-BOUND/runs/260605-gershgorin-reduction/result.json`.

5. Devil’s-advocate / self-adversarial review (CLAUDE.md 6.3)

- (α) *Attack the scope.* The result is confined to the scope stated in Sec. 1; a referee may push that boundary, which is fair — the package makes no claim outside it, and the precise scope is in the footer. In particular the package does NOT claim full STEP-5B closure, admissible-class exhaustiveness, or the full C_{full} theorem.
- (β) *Attack the tier / over-claim.* No prose here is stronger than the registered grade (footer no-overclaim line); the status is DRAFT ACCEPTED (T5/T6 support note), published-bundle confirmation HELD, and the claim-ledger tier is the arbiter.
- (γ) *Attack reproduction / self-containment.* Each reproduction script carries self-test asserts and ships in the folder’s bundle with its transitive dependencies (imports and runtime file reads, resolver v1.3.0); a referee runs them and diffs against `expected/`.

Quantitative sanity check (mandatory): the threshold is reproduced from its inputs, $((5972 - 8)/(4\sqrt{14}))^2 = 1.588 \times 10^5 \approx 1.59 \times 10^5$ (rounding consistent with the operator’s independent 1.58×10^5); the falsifier (footer) is concrete and pre-registered, so the claim is refutable by a single explicit construction; and the numerical values cited are reproduced by the folder’s script (192/192 asserts pass, exit 0, 2026-06-12 re-run).

6. Result footer

Result ID: B5-BEYOND-LAYER-BOUND / STEP-5B
rectangle-constant referee package

Precise statement: THEOREM (operator verdicts #9/#10, provenance: rectangle-constant-closure v1.3 S2-S3): for kappa-balanced patterns (H-KBAL: $A_{\max}^2 \leq \kappa^2 I/n$), $K(n) = \sum_{t \neq 0} w_t^2 / (\text{lam}' I)^2 \leq 8 + 4 \sqrt{14} \kappa^4 \sqrt{n}$; official threshold $n_{\text{adm}} \sim ((K_{\text{budget}} - 8) / (4 \sqrt{14}))^2 = ((5972 - 8) / (4 \sqrt{14}))^2 = 1.59e5$ at the anchor $I = 4e-4$. The 20/9 incidence route (2.2e10) is provisional (constant unpinned), not load-bearing.

Scope: the STEP-5B rectangle prefactor $K(n)$ and official threshold. Does NOT close full STEP-5B, admissible-class exhaustiveness, or full C_{full} .

Evidence grade: theorem-grade rectangle constant (T5/T6 support note); EXECUTED (192/192 asserts, exit 0).

Reproduction command: python codes/vacuum/beyond_layer_gershgorin_bound.py

Falsifier: a kappa-balanced configuration where $K(n)$ exceeds $8 + 4 \sqrt{14} \kappa^4 \sqrt{n}$; or the layer budget failing the 1.59e5 threshold.

Tier before / after: DRAFT ACCEPTED (operator verdict 2026-06-12); rectangle constant T5/T6 support note. No standalone tier promotion. Published-bundle confirmation HELD.

No-overclaim statement: the rectangle constant is theorem-grade within the H-KBAL scope; the 20/9 incidence route is provisional and not relied upon; full STEP-5B closure is not claimed.

Next required action: operator inspection + clean-run verification of beyond_layer_gershgorin_bound.py -> operator CONFIRM -> ONLY THEN package PUBLISHED bundle (policy sec.14).